

Worklet Details

1. Worklet ID: 24VI14MSRIT
2. College Name: MS Ramaiah Institute of Technology

KPIs achieved till now

- OpenCV implementation of Multiplication, Division, Normalisation and denormalisation along with NEON implementation.
- Image processing algorithm implementation using opencv - LIME, DUAL, WTHE, BM3D
- NEON Implementation of WTHE and LIME along with partial implementation of BM3D

Any Challenges/ Issues faced

- Implementation of Image processing techniques in NEON requires multiple algorithms to be implemented from scratch

Next Steps

- Evaluation of the NEON instructions for multiplication, division, normalisation and denormalisation and Image processing algorithms
- Implementation of another three papers in NEON

Key Achievements/ Outcome till now

- Understanding the NEON Instruction setting
- NEON Implementation of Multiplication, Division, Normalisation and denormalisation.
- NEON implementation of WTHE and LIME

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Image processing algorithms :

1. Weighted Threshold Histogram equalisation

Enhances the image contrast by modifying the image histogram by applying weights and thresholds

Weights are calculated based on probability mass function : If the PMF is lower than the threshold, it is set to zero, If it is above the threshold, it is set equal to the threshold.

Otherwise: It is equivalent to $((\text{pmf} - \text{low}) / (\text{high} - \text{low}))^r * \text{high}$



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Image processing algorithms :

2. BM3D (Block Matching and 3D Filtering)

Two step process applied sequentially. Each step includes 4 steps:

Block matching (Finding nearest similar blocks), Grouping (Grouping the similar blocks based on environment), Collaborative Filtering (Includes 3d transformation using hard thresholding and inverse transformation) and Aggregation (combining the filtered blocks to estimate the denoised image)



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Image processing algorithms :

3. LIME (Low light Image Enhancement)

It is enhancing the image quality using illumination maps. It tells about light condition of the blocks. The illumination map is refined and gamma correction is applied. This is followed by non linear operations for encoding and decoding the luminance (This adjusts the brightness).

The algorithm uses Gaussian kernel affinity to find nearest blocks and smoothen the illumination map. This corrects the under exposure.



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Image processing algorithms :

4. DUAL (Extension of LIME)

LIME algorithm only corrects the under exposure. DUAL algorithm uses the same method to correct the over exposure. It inverts the illumination map of the original image and applies the refinement on the inverse illumination map. After this the illumination map is inverted back to the original form. This corrects the over exposure.

DUAL algorithm corrects the under as well as the over exposure.

